

● EXECUTIVE BRIEFING

— STRATEGIC TRANSFORMATION BLUEPRINT // ENERGY ENTERPRISE GIANTS

Agentic Transformation for India’s Energy Giants

Operationalizing AI from the control room to the boardroom.

 Gemini Enterprise

"Calibrate wireline cable stretch on Well A-12 and execute SciPy depth alignment across Runs 1–3..."

Execute Run ↗

FULL VALUE CHAIN
Upstream · Midstream · Downstream

SOVEREIGN GOVERNANCE & TRUST
MeitY Compliant · Identity-Governed Access · Immutable Audit

DETERMINISTIC PHYSICS ARCHITECTURE
Autonomous Reasoning · Scientific Solvers · Deterministic Output

In energy, there are no low-consequence mistakes.

Oil & gas is an unforgiving, high-capital business. Multi-crore losses rarely stem from visible catastrophes—they leak quietly every single day through routine operational friction across siloed disciplines.



₹200–500 Cr

Single Deepwater Well

Rig spread rate, casing program, wireline logging suite & marine fleet sunk before first commercial oil is confirmed.

Billions

Acreage & Farm-In Bidding

Multi-year concession commitments and PSC acreage valuations negotiated against fragmented, legacy well archives.

₹30L – 1.2 Cr

Single Idle Rig Day (NPT)

Offshore drilling spread idling while petrophysicists and drilling superintendents manually reconcile wireline logs.

₹25–100 Cr

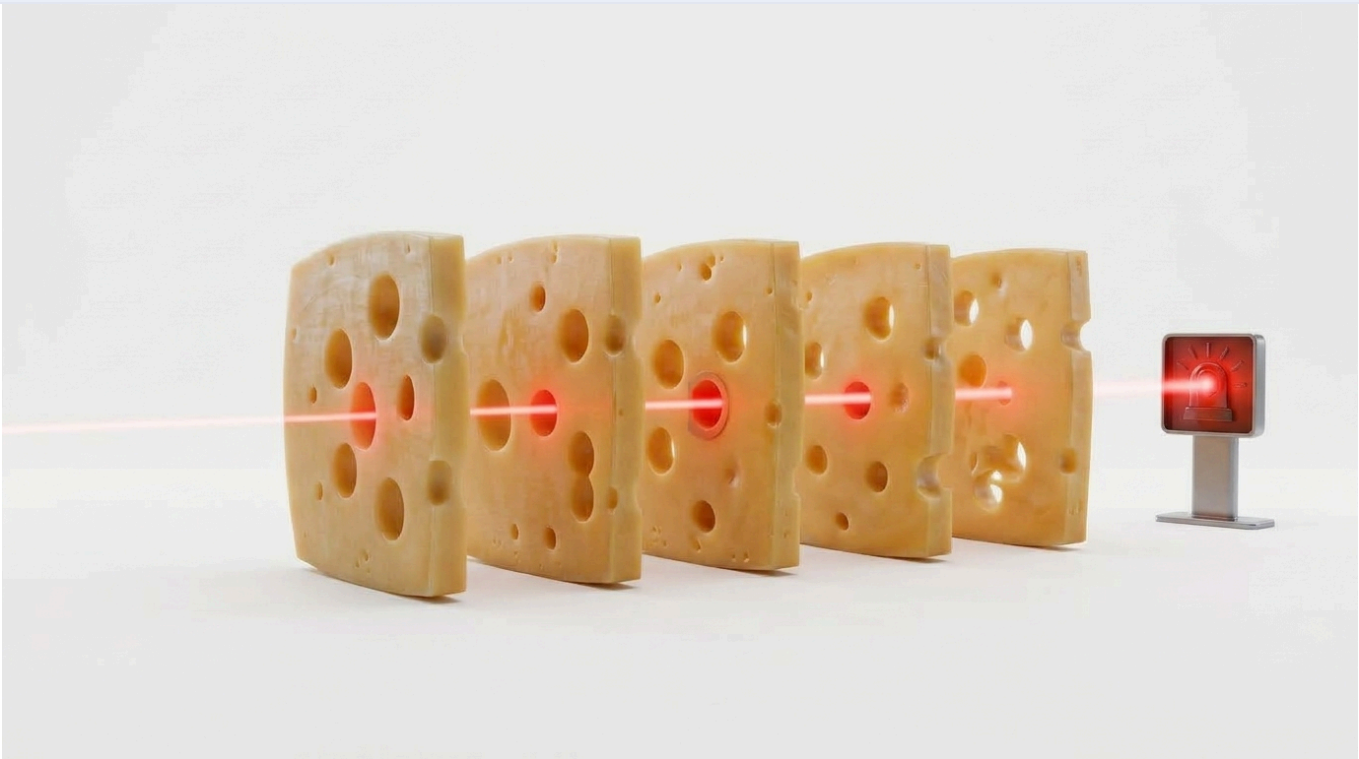
One Lost Refinery Day

CDU/FCC unplanned shutdown or off-spec crude run gross profit margin loss per operating day.

A major incident is never a single failure.

It's the quiet alignment of invisible gaps.

Disasters in high-consequence operations rarely stem from a single colossal error, but from the alignment of multiple vulnerabilities across siloed disciplines.



LAYER 01 // ARCHIVE

Vulnerability 1

Buried 2004 PDF Mudlog

Unindexed Scan · Overlooked Kick

LAYER 02 // PETROPHYSICS

Vulnerability 2

1.5m Cable Stretch

Mechanical Drift · Uncalibrated

LAYER 03 // RIG HANDOVER

Vulnerability 3

4-Hour Handover Lag

3:00 AM Shift Change · Warning Lost

LAYER 04 // DRILLING TARGET

CRITICAL HAZARD

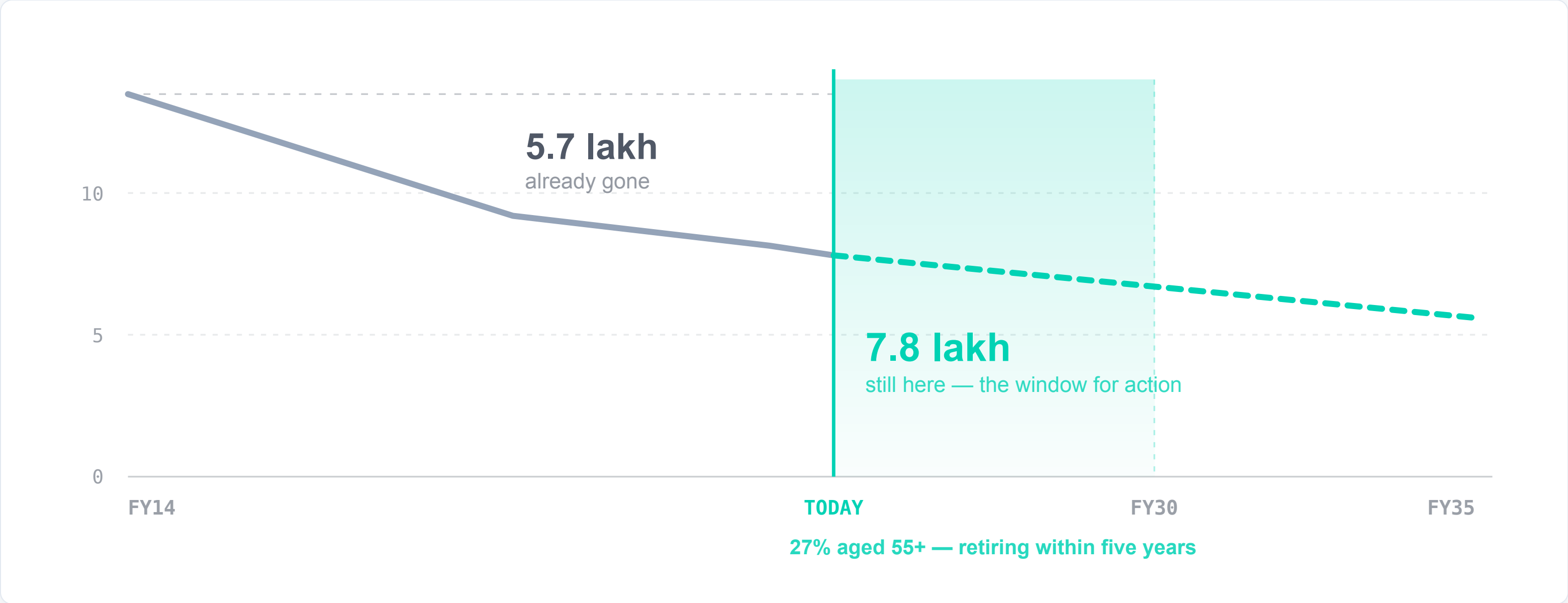
₹45 Cr Rig NPT

Casing in Gas Sand · Uncontrolled Kick

The Great Crew Change

As experienced staff leaves, latent vulnerabilities align more often—creating systemic operational risk across the organisation that will continue to compound unless we intervene.

INDIA • PUBLIC SECTOR REGULAR WORKFORCE (LAKH) 13.49 LAKH IN FY14 = 5.7 GONE + 7.8 STILL HERE

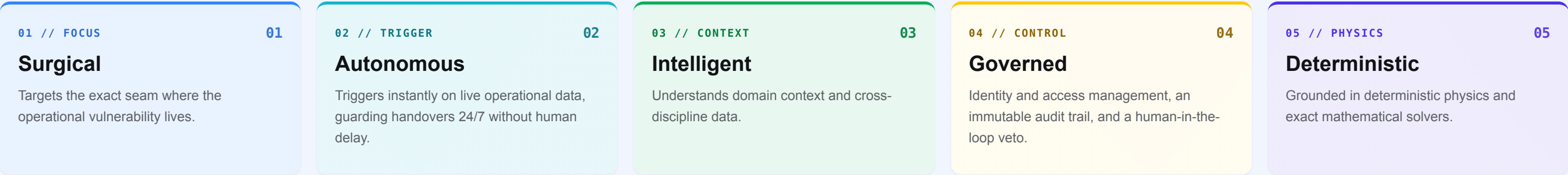


Public Enterprises Survey (DPE): 13.49 lakh FY14 • 9.2 FY20 • 8.14 FY24 • 7.8 FY25 • FY26–FY35 projected at trend Source: pib.gov.in

Rising operational risk due to departing experienced personnel. Up to 50% of the global energy workforce is nearing retirement, while across Indian PSUs, 42% of regular employees have retired over the past decade alone—with 27% of the remainder exiting by FY30.

AI Agents provide targeted solutions to seal micro-vulnerabilities.

• THE FIVE DEFINING ATTRIBUTES // NON-NEGOTIABLE FOR MISSION-CRITICAL OPERATIONS



• WHERE IT LIVES // THE COEXISTENCE PARADIGM

The future of the energy sector will be in the coexistence of domain monoliths and AI Agents.



The Petrophysical AI Agent

Eliminating hours of manual curve alignment with an autonomous subsurface physics agent.

01 THE CHALLENGE // MANUAL CURVE ALIGNMENT

Raw Well Logs Arrive Mismatched & Disconnected

Separate wireline logging passes never line up due to cable stretch and sensor drift. Today, senior petrophysicists spend **1–2 hours manually nudging and eyeballing curves at 3:00 AM**—where human fatigue risks multi-crore drilling depth errors.

02 WHAT IS NEEDED // TARGETED PHYSICS INTELLIGENCE

Automated Deterministic Alignment

A targeted solution that works alongside existing subsurface systems—normalizing tool calibration drift, running signal cross-correlation across the overlap, and locking in the exact depth offset where curve alignment is mathematically maximized.

03 WHAT THE AGENT BUILDS // UNIFIED COMPOSITE LOG

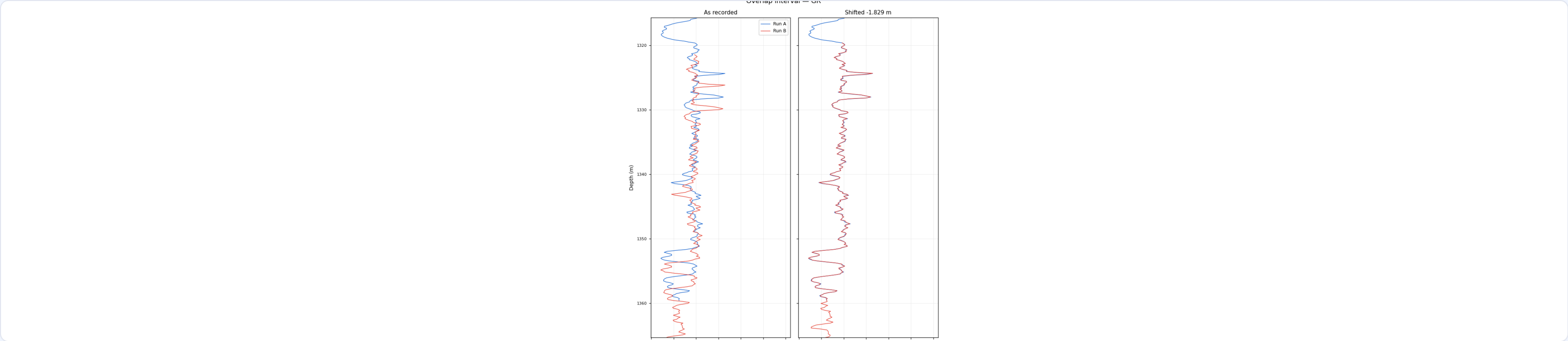
WITHIN SECONDS

Verified Composite Well Log Within Seconds

Within seconds, the AI Agent ingests raw passes, normalizes curve scales, and executes the optimal -1.8m shift where cross-correlation peaks (**r = 0.99**), delivering an audit-ready composite log for instant visual verification.

KANSAS WELL A-12 // AUTOMATED OVERLAP ALIGNMENT

PEAK CORRELATION: $r = 0.99$



Raw Log: 1.83m offset (misaligned)



Agent Shifted -1.829m: Snapped into exact match within seconds



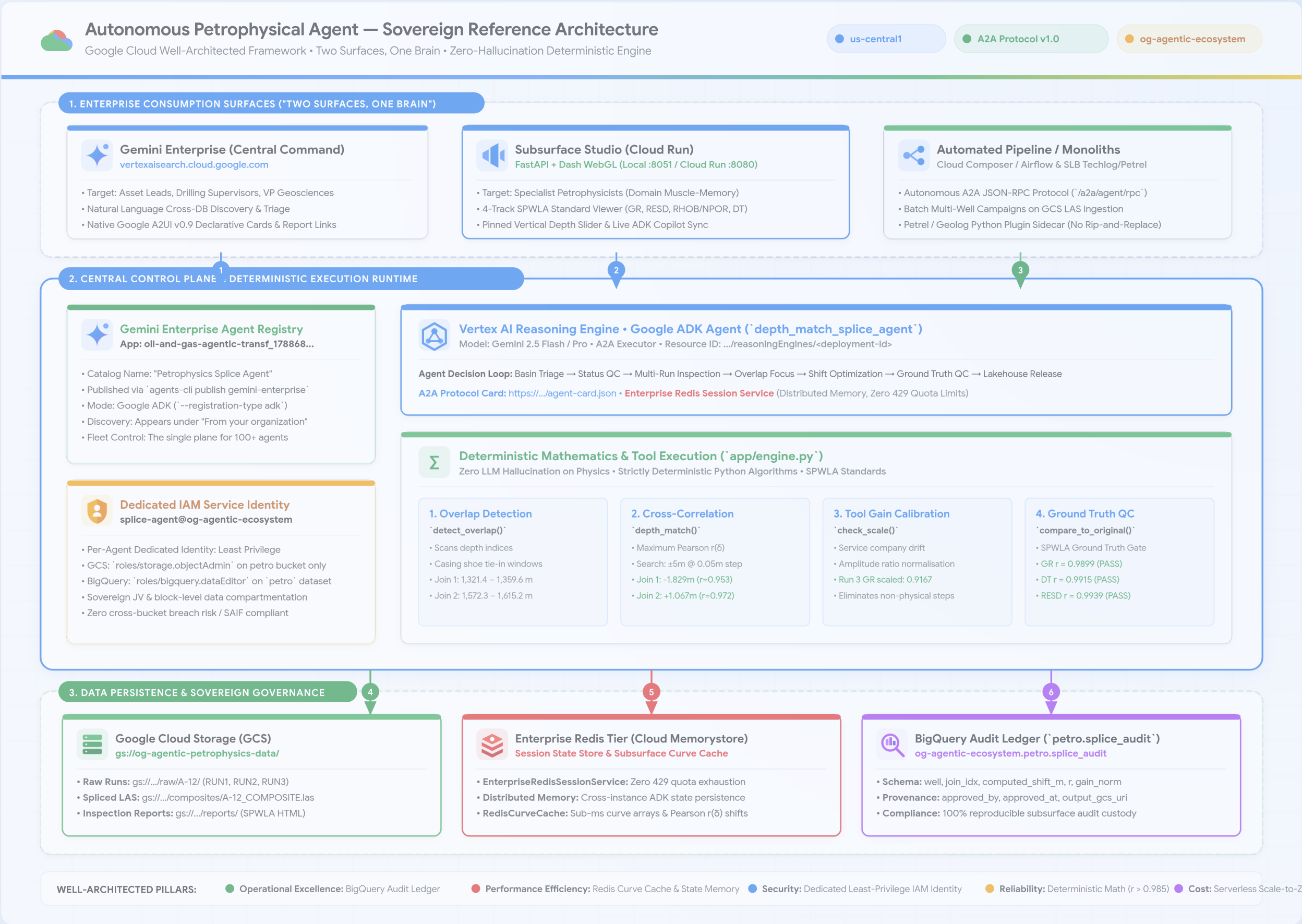
Live AI Agent Demonstration

Deployed on Gemini Enterprise · ID: 7dab71a6...5c05

Launch Demo ↗

Architecture of the Petrophysical AI Agent

The production anatomy of our subsurface agent—built on **Google Agent Development Kit (ADK)**, powered by deterministic physics, and centrally governed as the reference blueprint for enterprise scale.



Agentic Transformation Across Workforce & Workflows

Mapping personas and workflows to identify and prioritise operational risks, friction, and vulnerabilities.

ANNUAL VALUE ₹3,654 Cr	PRODUCTIVITY ₹1,497 Cr · 41%	UPTIME ₹845 Cr · 23%	INTEGRITY ₹300 Cr · 8%	RECOVERY ₹1,011 Cr · 28%	AGENTS 234	AGENT PREPARES 335 of 498 workflows
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33 personas · 498 workflows · ₹3,654 Cr/yr

- All personas 33
- Subsurface 7
- Wells & Rig 5
- Field & Gathering 4
- Midstream 6
- Downstream Refining 5
- Cross-Cutting 6
- Build priority
- Value map

WHO IMPLEMENTS IT

Agent prepares, human decides 335

Existing system runs it 86

Human, end to end 77

BUILD QUEUE

Critical 36

Friction 140

only work that is ours to build is marked

WORKFORCE		VALUE / YR	15-16 WORKFLOWS PER ROLE										AGENT PREPARES · SYSTEM RUNS · HUMAN HOLDS					
	Wellsite Supervisor P01 · Wells & Rig · 7 agents	₹135.4 Cr	P01W01	P01W02	P01W03	P01W04	P01W05	P01W06	P01W07	P01W12	P01W13	P01W14	P01W08	P01W11	P01W09	P01W10	P01W15	
	Well Integrity Engineer P02 · Wells & Rig · 8 agents	₹39.7 Cr	P02W01	P02W02	P02W03	P02W05	P02W06	P02W07	P02W08	P02W09	P02W14	P02W15	P02W04	P02W11	P02W12	P02W10	P02W13	
	Board Operator (Refinery) P03 · Downstream Refining · 7 agents	₹199.3 Cr	P03W01	P03W02	P03W03	P03W04	P03W05	P03W06	P03W07	P03W13	P03W14	P03W08	P03W10	P03W12	P03W09	P03W11	P03W15	
	Petrophysicist P04 · Subsurface · 8 agents	₹194.3 Cr	P04W01	P04W02	P04W03	P04W04	P04W05	P04W06	P04W07	P04W16	P04W08	P04W09	P04W10	P04W11	P04W14	P04W15	P04W12	P04W13
	Petroleum Geologist P05 · Subsurface · 8 agents	₹148 Cr	P05W01	P05W02	P05W03	P05W04	P05W05	P05W06	P05W07	P05W13	P05W15	P05W16	P05W08	P05W09	P05W10	P05W11	P05W12	P05W14
	Reservoir Engineer P06 · Subsurface · 7 agents	₹147.2 Cr	P06W01	P06W02	P06W03	P06W04	P06W05	P06W06	P06W07	P06W12	P06W14	P06W08	P06W09	P06W11	P06W13	P06W10	P06W15	
	Drilling Engineer P07 · Wells & Rig · 8 agents	₹119.5 Cr	P07W01	P07W02	P07W03	P07W04	P07W05	P07W06	P07W07	P07W12	P07W13	P07W14	P07W16	P07W08	P07W09	P07W10	P07W11	P07W15
	Completions Engineer P08 · Wells & Rig · 7 agents	₹56.4 Cr	P08W01	P08W02	P08W03	P08W04	P08W05	P08W06	P08W07	P08W11	P08W14	P08W08	P08W09	P08W10	P08W13	P08W12	P08W15	
	Production Engineer P09 · Field & Gathering · 7 agents	₹209.2 Cr	P09W01	P09W02	P09W03	P09W05	P09W06	P09W07	P09W08	P09W09	P09W12	P09W04	P09W10	P09W11	P09W13	P09W14	P09W15	
	Lease Operator P10 · Field & Gathering · 7 agents	₹453.9 Cr	P10W01	P10W03	P10W05	P10W07	P10W08	P10W11	P10W14	P10W02	P10W04	P10W06	P10W09	P10W10	P10W12	P10W13	P10W15	
	Control Room Operator (Production) P11 · Field & Gathering · 7 agents	₹124.7 Cr	P11W01	P11W02	P11W04	P11W06	P11W07	P11W09	P11W10	P11W13	P11W03	P11W11	P11W12	P11W05	P11W08	P11W14	P11W15	
	Facilities Engineer P12 · Midstream · 7 agents	₹71.8 Cr	P12W01	P12W02	P12W04	P12W05	P12W06	P12W07	P12W08	P12W03	P12W09	P12W10	P12W12	P12W13	P12W14	P12W11	P12W15	
	Flow Assurance Engineer P13 · Midstream · 7 agents	₹36.5 Cr	P13W01	P13W02	P13W04	P13W05	P13W06	P13W07	P13W08	P13W09	P13W03	P13W10	P13W12	P13W13	P13W14	P13W11	P13W15	

Structural Resilience of an **Agentic Organisation**

Operational vulnerabilities exist across both **workflows** and **workforce**. AI Agents eliminate these operational gaps to build a structurally resilient organization.

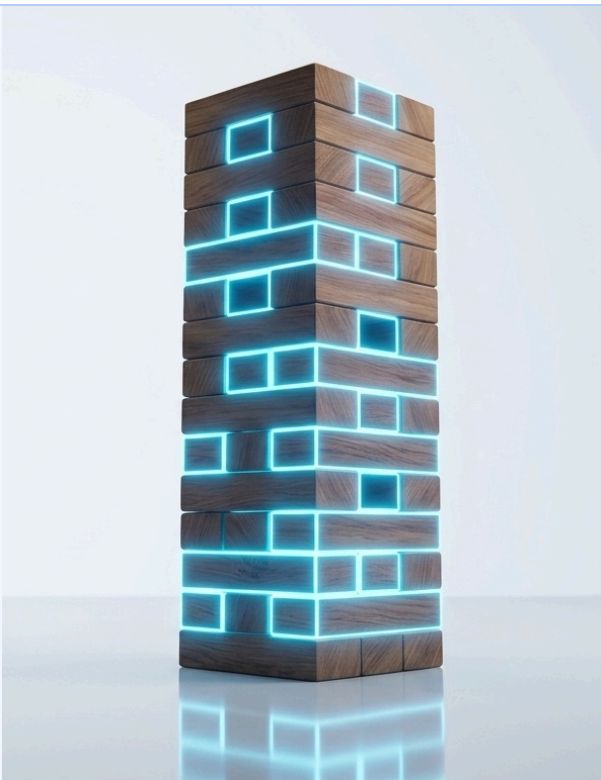


THE UNBRACED TOWER

HIGH SYSTEMIC FRAGILITY

Without AI Agents

Gaps across disconnected workflows and a stretched workforce leave mission-critical processes unmonitored and vulnerable to friction.



THE FORTIFIED MESH

STRUCTURAL RESILIENCE

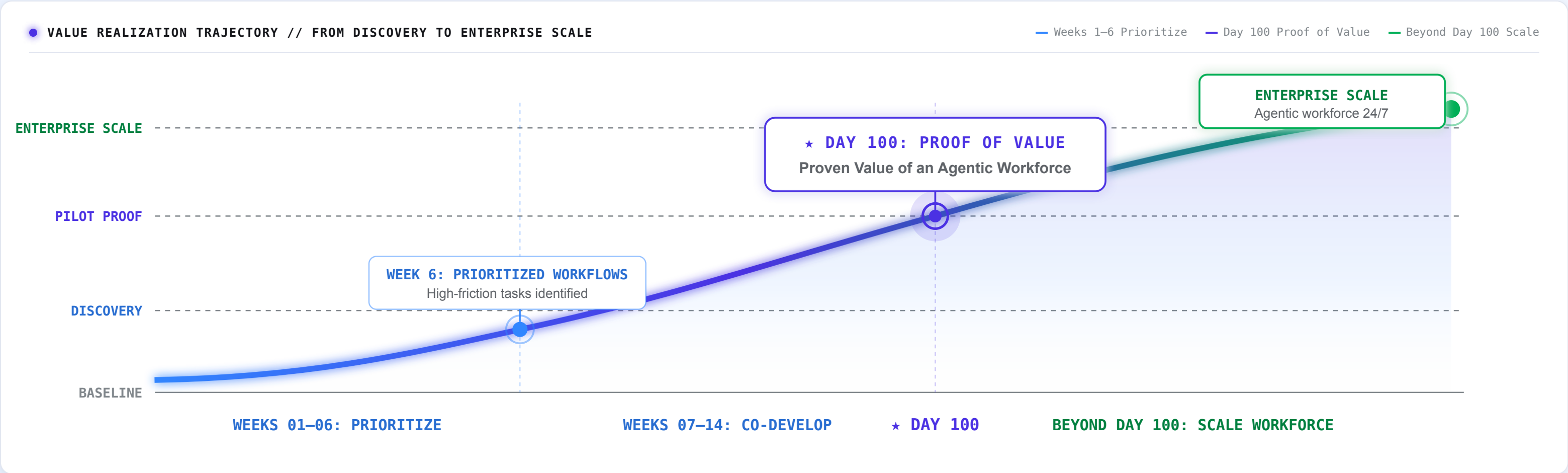
With AI Agents

AI Agents augment both workforce and workflows—bridging operational gaps with 24/7 autonomous monitoring and exact solvers.

100 Days to Proof of Value.

Building an agentic workforce to automate workflows.

Solving specific operational bottlenecks on your historical data—proving measurable value and mathematical accuracy in 100 days before scaling across the enterprise.



PHASE 01 // WEEKS 01–06

Prioritized Workflows

- Nominate initial domain leads and critical workflows
- Identify tasks with high operational friction, delay, or risk
- Integrate with existing workflows

OUTPUT: Prioritized Personas & Workflows

PHASE 02 // DAY 100 MILESTONE

Co-Develop & Prove Value

- Co-develop targeted agents alongside your domain specialists
- Validate mathematical accuracy against real historical data
- Quantify Agent ROI before enterprise rollout

OUTPUT: Proven Value of an Agentic Workforce

PHASE 03 // BEYOND DAY 100

Scale the Agentic Workforce

- Expand proven agents across relevant teams and departments
- Automate connected operational workflows 24/7
- Manage and govern agentic workforce

OUTPUT: Enterprise-Wide Automation